CONTINUATION.md — vasic-digital tmux

Last updated: 2026-07-22T19:05:00Z

§0 — How to resume work in any CLI agent

Paste this prompt:

> Read `CONTINUATION.md` at the repo root. Identify the topmost item under `§3 Active work` with status IN PROGRESS or BLOCKED. Re-read `Constitution.md`, `CLAUDE.md`, `AGENTS.md`, `Issues.md`, `Fixed.md` for mandates and current backlog. Resume from current state. Update this document as you work.

§1 — Snapshot

Field	Value
Repo	vasic-digital/tmux on GitHub + GitLab
Origin	Migrated from ATMOSphere project (`scripts/tmux/`, `docker/Dockerfile.tmux-build`, `docs/guides/TMUX_OPTIMIZED_BUILD.md`) on 2026-05-07
Pinned tmux	upstream tag `3.6a`
Version	**1.0.37 prepared** (versionCode 38; release tag pending operator gate) — host-adaptive CPUQuota default (cores × 15%, floor 200%) replacing the fixed 200% that was the proven root cause of progressive session sluggishness on many-core hosts (evidence `docs/qa/cpu-throttle-20260722/`; test 86 + meta-mutation M-CPUADAPT). Previous: v1.0.36 (2026-07-17, extended-keys-format csi-u), v1.0.35 (session-name sanitization).
Verification (this cycle)	`setup.sh` v1.0.35: PASS=69 FAIL=0 SKIP=13 (nezha, Linux x86_64). `run_all.sh` v1.0.35: PASS=69 FAIL=0

Field	Value
	SKIP=13. `meta_test_false_positive_proof.sh`: 66 mutations caught, 0 escaped, 11 skipped (M82 skipped by automated harness; manually reproduced and proven to catch). Live retest: `tmx new -s 'hello world' -d` → `hello-world`; `tmx new -s '  messy! name  ' -d` → `messy-name`; `tmx new -s 'hello world:red' -d` → `hello-world` with `bg=red`. Release tag v1.0.35 pending commit + publish.
Governance docs	`constitution/` submodule (HelixConstitution, pinned `1576d3d`); `Containers/` submodule (pinned `fbef9d6`); project `Constitution.md` (Project Articles §101–§170 extends the submodule), `CLAUDE.md`, `AGENTS.md`, `QWEN.md`, `GEMINI.md` (all lockstep to §11.4.170), `Issues.md`, `Fixed.md`, this document

§2 — Mandates (canonical authority)

- `Constitution.md` §1 anti-bluff covenant (verbatim user-mandate quote propagated)
- `Constitution.md` §11.4.1 FAIL-bluffs equally forbidden
- `Constitution.md` §11.4.2 recorded-evidence requirement
- `Constitution.md` §11.4.3 per-host-topology test dispatch
- `Constitution.md` §11.4.4 test-interrupt-on-discovery + 4-layer test coverage
- `Constitution.md` §11.4.5 audio + video quality analysis (N/A for tmux scope; principle generalized)
- `Constitution.md` §11.4.6 — No-guessing mandate (verbatim user-mandate quote)
- `Constitution.md` §9 absolute data safety
- `Constitution.md` §12.6 60% host memory budget (per-session `TMX_MEM` is the enforcement)
- `Constitution.md` §5 / §12.10 continuation-document sacred invariant

§3 — Active work

§3.29 — Progressive-sluggishness root cause fixed: host-adaptive CPUQuota (2026-07-22) — v1.0.37 prepared

Status: READY FOR RELEASE GATE — fix (adaptive quota + quota-sized `cpu.max.burst` bank) + test 86 (G1–G6) + M-CPUADAPT mutation + captured evidence committed; release tag v1.0.37 (versionCode 38) awaits operator/conductor approval. Live forensics (coordinator measurement arm, 2026-07-22) confirmed the throttle mechanism as the dominant cause: even at 960% quota, 16.9% of periods throttled during fleet bursts (avg demand 4.35 CPUs) — the burst bank closes that (A/B: 33% → 0% throttled on identical spiky workload). OPEN follow-up (operator decision): split the interactive tmux server into its own scope, separate from the subagent fleet (changes crash-isolation topology). Root cause: fixed `CPUQuota=200%` per `tmx` scope throttled the whole session tree (tmux server included) on many-core hosts — cgroup `cpu.stat` showed 18.8% throttled periods / 1757 s forced idle; typing echo + timer redraws froze for the rest of every 100 ms CFS period. Fix: `_default_cpu_pct()` in `scripts/tmx.template` (cores × 15%, floor 200%, cap cores × 100; `TMX_CPU` override preserved; `TMX_CPU=auto` = adaptive). Evidence: `docs/qa/cpu-throttle-20260722/` (before/after throttle probes, latency series, 45×10 s cgroup time-series, test-86 polarity matrix). Live production scope remediated in place (`cpu.max=960000`). Config-layer latency settings audited already-optimal (escape-time 0, no `#()` in status) — no further change without new measured evidence.

§3.28 — Wizard + session-password redesign (2026-07-05) — docs landed, code in sibling tasks

Status: IN PROGRESS — Task 11 of 14 (documentation) landing this commit.

Plan: `docs/superpowers/plans/2026-07-05-tmx-wizard-password-redesign.md` **Spec:** `docs/superpowers/specs/2026-07-05-tmx-wizard-password-redesign-design.md`

Four user-visible requirements (operator mandate 2026-07-05), tracked as new OPEN workable items in the SQLite SSoT + `Issues.md` §G:

- **TMX-072** (Feature) — wizard-created sessions get a random 4-digit name suffix (`name-NNNN`; `TMX_EXACT_NAME=1` opts out).
- **TMX-073** (Feature) — password input masked with `*` while typing.

- **TMX-074** (Bug) — reopening a password-protected (idle-recycled) session no longer asks for the password twice; verify-once.
- **TMX-075** (Feature) — wizard offers a numbered picker of existing sessions (`+ 0) None`) on blank input.

This task (Task 11 — documentation) delivered: README What's-new + docs-map rows; `docs/guides/tmx-shell-integration.md` wizard section rewritten (suffix, picker, masking, `TMX_EXACT_NAME`); new `docs/guides/tmx-session-passwords.md` standalone guide + Mermaid decision-flow diagram (rendered PNG, confirmed image-not-raw-source in the PDF per §11.4.168); new `docs/guides/FAQ.md`; the 4 workable items added to the SQLite SSoT + `Issues.md` §G regenerated byte-identical (`diff` clean); all touched Markdown re-exported to HTML/PDF/DOCX.

Next (remaining plan tasks, concurrent): the CODE tasks — `has-password` Go subcommand, `_read_password_masked` + restructured `new/attach` in `scripts/tmx.template`, the always-create + picker wizard in `scripts/tmx-shell-init.sh.template`, and tests 77–84 + 66/68 reconciliation — then full-suite retest + a validated release tag per §11.4.40 / §11.4.126. Terminal goal: a new fully-validated version published to all remotes.

Note (§11.4.6): `workable-items validate` reports a single pre-existing `§11.4.54 ticket sequence gap: expected TMX-071, found TMX-072` — TMX-071 was a previously-reverted item and the `next_atm_id` counter was already at 72, so the new items correctly start there. NOT to be back-filled or renumbered.

§3.27 — v1.0.26 RELEASED (per-session color) + BOTH HOSTS INSTALLED → 2026-06-19

Status: RELEASED v1.0.26 / versionCode 27 — tag `v1.0.26` at `5244f4e` (`13ef83b`), published GitHub (https://github.com/vasic-digital/tmux/releases/tag/v1.0.26) + GitLab (https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.26). main HEAD `5244f4e` identical on local + github + gitlab.

Feature: operator-chosen per-session tmux color via `tmx new -s name:color[:ignored]`. Color validated (tmux names / `colour0-255` / `#hex`), applied to all 4 "green" surfaces (status-bar bg, active-pane border, clock-mode-colour, current-window bg), persisted in `~/.tmx/state.json` (schema 1→2 additive `color` field), re-used on bare-name re-runs. Precedence: inline > persisted > hostname > default-green. Escaped `\:` in name; extra `:fields` ignored. tmx-state-bin `v1.0.9 →`

`v1.1.0` (`set-color`/`get-color` + `list` 4th col). Invalid color rejected (exit 5) before any session created. macOS bridge unchanged → §11.4.81 parity by construction.

Anti-bluff evidence (§11.4.5/§11.4.69):

- `scripts/tests/63_session_color.sh` — 8/8 PASS, every assertion reads LIVE `show-options` from the real server. Deterministic 3× (§11.4.50).
- `scripts/tests/64_session_color_parse_unit.sh` — 17/17 pure parser/validation unit tests incl. bash↔Go canonical-name-list parity.
- §1.1 paired mutations M25 (strip `_apply_color`) + M26 (neutralize `set-color`) both CAUGHT by test 63.
- HelixQA Challenge TMUX-CH-53 (blocker).
- **Mistborn INSTALLED** (`setup.sh --rebuild`): native Darwin arm64 build, tmux 3.6a, tmx-state-bin v1.1.0, VERSION 1.0.26. Verify gate `PASS=57 FAIL=0 SKIP=6` (test 63 = 8/8, test 64 = 17/17 ran during install). **Live end-user proof on the installed build:** `tmx new -s proofofwork:red` → `status-style=bg=red`, `clock-mode-colour=red`, `pane-active-border-style=fg=red`; persisted (`get-color` → `red`) + bare re-run reuses it; `#hex` works (`bg=#e63946`); invalid `notacolor` rejected exit 5 + no server.

** nezha — INSTALLED** (`/home/milosvasic/Projects/tmux`, 2026-06-19): after the operator confirmed nezha back online (§11.4.144 unblock condition met — `ssh ... nezha.local` returned `Linux x86_64`), clean fast-forward `05569e5 (v1.0.25)` → `e1a2530 (v1.0.26)` (local in-flight D2 TMPDIR edits stashed — already superseded upstream at `5f86936`), `setup.sh --rebuild` → Linux ELF tmux 3.6a + `tmx-state-bin v1.1.0`. Verify gate **PASS=49 FAIL=0 SKIP=14**; test 63 **8/8** (T8 hostname fallback → `bg=colour94`, matching Mistborn → cross-OS parity §11.4.81) + test 64 **17/17**. Live color proof on nezha: `tmx new -s nezproof:red` → all 4 surfaces red; persisted; `#hex`; invalid rejected exit 5. `NEZHA-INSTALL-v1.0.26-001` closed → Fixed.md. **Both hosts now at v1.0.26, clean + verified.**

§11.4.151 release-prefix — NOT adopted this release: project tag history is `v1.0.6`…`v1.0.25` (no prefix); operator chose `v1.0.26` to keep the series continuous (decision recorded). The universal `tmux-<ver>` prefix remains a standing operator decision for a future major release boundary.

Known / tracked separately (not regressions): `M24-ESCAPE-001` NOW FIXED (v1.0.27, meta-test CAUGHT 37/0/17). Test 13 Darwin `ulimit -u` honest-gap (§11.4.81, pre-existing).

§3.28 — v1.0.27 RELEASED: name:color prompt rejection fixed + Issues.md burn-down (M24/A2/D2 all closed) → 2026-06-19

Status: RELEASED v1.0.27 / versionCode 28 — tag `v1.0.27` pushed + published GitHub (https://github.com/vasic-digital/tmux/releases/tag/v1.0.27) + GitLab (https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.27). Both hosts validated:

- Mistborn (macOS arm64): 58 PASS / 0 FAIL / 6 SKIP + meta 37 CAUGHT / 0 ESCAPED / 17 SKIPPED
- Nezha (Linux x86_64): 49 PASS / 0 FAIL / 14 SKIP (meta pending post-push pull)
- commit `67d5f01` at HEAD + tag `v1.0.27` — github + gitlab identical.

Operator report (2026-06-19) — §11.4.138 operator-escape:

```
[tmx] invalid session name 'home:red'; allowed: [A-Za-z0-9_.-]
{1,64}
```

ROOT CAUSE (Phase 1 complete, fix applied). The `tmx` wrapper's `_parse_session_value()` already handled `NAME:color` correctly — the gap was the shell-init prompt AND SSH-dispatcher running BEFORE the wrapper with `[A-Za-z0-9_.-]` char-set + 64-char limit (a textbook §11.4.108 SOURCE→USER-VISIBLE gap: GREEN suite, broken-for-user feature). Test 65 existed for this exact bug (written as the anti-bluff regression guard for v1.0.26) but the underlying templates had not been updated.

Fix applied (all 4 changes reconciled per §11.4.120):

- `scripts/tmx-shell-init.sh.template` — char-set `[A-Za-z0-9_#.:-]`, max 64→80
- `scripts/tmx-ssh-dispatch.sh.template` — same fix
- `scripts/tests/65_shell_init_color_prompt.sh` — updated T2/T3 for full-spec forwarding (not pre-parsed)
- `scripts/tests/35_session_name_validation.sh` — LONG65→LONG81, limit 64→80

Verification: 58 PASS / 0 FAIL / 6 SKIP (63=8/8, 64=17/17, 65=6/6). Meta 35/35 caught.

§3.28.1 — M24-ESCAPE-001 (Stream 1) → DONE. Root cause: M24 mutation used `re.subn(pat, '', src, count=1)` which matched `_apply_color()` first (wrong function), while test 26 exercises `_apply_host_color()`. Fixed by removing `count=1` → strips from BOTH functions. Meta-test before: 34 CAUGHT / 3 ESCAPED (M24 was 1 of 3). After: **37 CAUGHT / 0 ESCAPED / 17 SKIPPED**. M24 verified: stripped 6 set-lines across both color functions → test 26 FAILs (T2/T3/T4), reverted → PASSes.

§3.28.2 — A2 RUNALL-NATIVE-RESOLVE-001 (Stream 2) → DONE. `run_all.sh` now OS-aware: builds `TMUX_BIN_DEFAULT="$REPO_ROOT/tmux/build-darwin/bin/tmux"` with fallback to `tmux/build/bin/tmux`, respects `$TMUX_BIN` override. Gate `CM-RUNALL-OS-AWARE` PASSes.

§3.28.3 — D2 TMPDIR-HARDCODE-001 (Stream 3) → DONE. 35 test files updated: bare `/tmp` paths replaced with `"$SCRATCH/..."` where `SCRATCH="${TMPDIR:-/tmp}"`, plus writability preflight guard per §11.4.3. Gate `CM-NO-HARDCODED-TMP-SCRATCH` PASSes (16 cwd/state/dispatch tests).

QA: `docs/qa/2026-06-19-v1.0.27-burndown/`.

Anti-bluff verification (current host — macOS Mistborn):

- Full suite: **58 PASS / 0 FAIL / 6 SKIP** (test 63=8/8, 64=17/17, 65=6/6)
- Meta-test: **37 CAUGHT / 0 ESCAPED / 17 SKIPPED**
- `bash scripts/setup.sh` — GREEN, installed.
- `bash scripts/setup.sh --verify-only` — GREEN.

Nezha deploy: completed 2026-06-19 post-tag (verified 49/0/14 + meta 37/0/17).

§3.29 ✅ — v1.0.28 RELEASED: per-session password protection + governance sync (§11.4.159–170) → 2026-06-28

Status: RELEASED v1.0.28 / versionCode 29 — tag `v1.0.28` at `f30cd8b`, published GitHub (https://github.com/vasic-digital/tmux/releases/tag/v1.0.28) + GitLab (https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.28).

What landed:

- **Per-session password protection** — `tmx new -s NAME` prompts for optional password; `tmx attach -t NAME` verifies before granting access. SHA-256 hashed storage, schema v3, graceful degradation.

- **tmx-state-bin v1.2.0** — new `set-password` / `verify-password` subcommands.
- **Test 66** — 5 invariants × 3 iterations (15/15 PASS).
- **Meta-test M27** — `verifyPassword` always-true mutation CAUGHT by test 66 T3.
- **Governance sync** — constitution `1d408cb` → `1576d3d`; §11.4.159–§11.4.170 propagated to all 5 governance carriers.

Verification: Full suite PASS=51/FAIL=0/SKIP=14. Meta-test 55 CAUGHT/0 ESCAPED/8 SKIPPED.

Pending: Release tag, deploy on Mistborn + mistborn.local.

§3.26 — v1.0.25 RELEASED (test fixes + submodule currency + governance cascade) → 2026-06-16

Status: RELEASED v1.0.25 / versionCode 26 — tag at `3dd6a7c`, published GitHub (https://github.com/vasic-digital/tmux/releases/tag/v1.0.25) + GitLab (https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.25). Both hosts (macOS Mistborn

- nezha) synced to `05569e5` (= tag + governance cascade), VERSION 1.0.25, tmux 3.6a, setup + smoke (tmx new/ls/kill) GREEN. Authoritative validation: nezha rel25c destructive gate verify PASS=50 FAIL=0 SKIP=11 (V=0) + meta 52 CAUGHT/0 ESCAPED; macOS setup run_all 55/0/6. Test 17 determinism: macOS 5/5 + nezha 20/20 under full all-core CPU saturation. **Governance cascade (`05569e5`):** CLAUDE/AGENTS/QWEN appended §11.4.101–158 verbatim + **GEMINI.md created** (§11.4.157 lockstep, byte-identical to CLAUDE.md); §11.4.65 siblings regenerated. **Operator-pending:** §11.4.151 release-prefix (`tmux-<ver>`) — NOT adopted mid-series (project uses `vN.N.N`); decide before next release. Both hosts at latest, clean.

(Historical detail of the cycle follows.) Per operator "always target latest of all submodules": advanced HelixConstitution `3f4d690` → `1d408cb` (+67 commits, §11.4.140–158); Containers already at latest `18ed03d`. Functional inheritance verified (test 18 PASS=10/0); release gate (verify.sh §11.4.87..99) unaffected. Host-level: codegraph macOS skew fixed (stale `~/.local/bin` symlink → both hosts 1.0.1); 668 dead macOS test sockets removed (678→10, no live session touched).

Test 17 flake — root-caused + fixed (A49, 2 vectors). `17_scrollback_copy_mode.sh` flaked under heavy host load (§11.4.1 FAIL-bluffs, NOT a product defect — scrollback works for the end user).

Vector 1: the T4 generator spelled `SCROLLMARK_FIRST`/`SCROLLMARK_LAST` inline, so the terminal's echo of the TYPED command line contained `SCROLLMARK_LAST` and the `GEN_OK` poll matched the command echo before any real output (forced-slow probe on nezha: command-echo grep=1, real-output grep=0). Fixed by assembling markers from a shell variable (`M=SCROLLMARK; echo ${M}_LAST`) → every grep matches REAL OUTPUT only. Vector 2: under all-core CPU saturation a `capture-pane` grid dump can momentarily lack the oldest `FIRST` marker although the buffer genuinely retains all 3000 lines (captured: 3006 lines, LAST present, 2998 numbered markers, history-limit 50000). Fixed by asserting retention via the **race-free `#{history_size}` counter** (`>= 2900`, `display-message -p`) at T4.2, not a grid dump (the no-eviction value is a deterministic ~2982; the OLD 2000 cap pins it at 2000 and evicts line 1) — T4.3/T4.4 (operator copy-mode reach of FIRST) already proved retention robustly and never raced. Test source only (`scripts/tests/17_scrollback_copy_mode.sh`); assertions of the end-user guarantee unchanged. **Determinism (§11.4.50) — PROVEN:** macOS N=5 GREEN (5/5) + nezha N=20 under full all-core CPU saturation GREEN (20/20); `history_size` identical every run (macOS 2981 / nezha 2982 — counter does not race). Pending before tag: the §11.4.40 full destructive gate (rel25c, running) GREEN.

Known follow-up (non-blocking): verbatim/cascade mirroring of §11.4.140–158 into project CLAUDE.md/AGENTS.md/QWEN.md (for non-`@import` agents) — functional inheritance already works via the import; release gate does not enforce these. Operator-pending on nezha: `sudo bash scripts/build_oom_set.sh --install` + `sudo apt-get install -y stress-ng` to promote tests 08/12/14 from SKIP to PASS.

§3.25 — v1.0.24 RELEASE (distribution orchestrator) → 2026-06-16

Status: RELEASING v1.0.24 / versionCode 25. Tooling + library-hardening release (tmux product surface unchanged from v1.0.23). Bundles §3.24 (tmx-orchestrator + Containers fixes) under a tagged version published to GitHub + GitLab. Release gate green: nezha verify.sh PASS=50/0/11 + meta 52 CAUGHT/0 ESCAPED; nezha Containers full suite (unit + integration vs real podman) all 40 pkgs ok; macOS run_all 55/0/6; adversarial code-review GO. Containers pointer 61e01dc → 18ed03d (pushed github+gitlab).

§3.24 — Distribution orchestrator binary (Containers submodule) + nezha as heavy-test host → 2026-06-16

Status: DONE — orchestrator working, validated on nezha, committed.

What landed (A48). Operator mandate: "Make a proper binary using the Containers submodule lib to orchestrate the distribution — for when/if we need it for our testing needs." Delivered `scripts/tmx-orchestrator/` (Go consumer of the decoupled `digital.vasic.containers` submodule via `replace`): `hosts` / `distribute` / `down`. nezha.local registered in `Containers/.env` (gitignored). Extended the Containers library (§11.4.76) with `ContainerRequirements.Ports`

- `-p` rendering (`1b9da9b`) and a command-injection allowlist fix from the automated security review (`82bd586`) — both pushed github+gitlab; tmux pointer 61e01dc → 82bd586.

Validation (captured — `docs/qa/2026-06-16-tmx-orchestrator/evidence.md`): macOS conductor → nezha over SSH+podman: `hosts` real /proc resources; `distribute` → real nginx on nezha `0.0.0.0:18080->80/tcp`, HTTP health 200 (podman ps + curl confirm); `down` → REMOVED-CLEAN. Runtime test `scripts/tests/62_distribution_orchestrator.sh` PASS=4/0 (opt-in TMX_TEST_REMOTE=1). Containers full unit suite on nezha all 37 pkgs ok. Two real bluff-audits (feature + security guard). Containers submodule was NOT modified for tmux-specifics (CONST-051).

Post-landing validation + hardening (2026-06-16, parallel-stream round): Dual-host green — macOS full suite PASS=55/0/6 (zero regression; +1 SKIP = new test 62), nezha Containers integration tests (`-tags=integration`) all 40 pkgs `ok` against real podman. Orchestrator hardened: `distribute` now prints a cleanup hint when the health check fails (container left running for inspection, no silent leak — verified via a real induced HTTP failure). README documents the `--health tcp` vs `http` distinction (TCP connect-only false-passes on a published port; HTTP forwards through — verified). §11.4.65 .html/.pdf siblings generated for the new/changed docs. nezha synced to tmux a130a8f + Containers 82bd586.

Adversarial code-review round (§11.4.125/§11.4.134 review-until-GO, 2026-06-16). A dispatched review found 2 major + minors; all fixed with real evidence:

- **MAJOR (lib socket leak):** `SSHExecutor.Execute` released the pooled ControlMaster ref only on success → leaked on error paths. Fixed (defer release on all paths) in Containers `a856865` + white-box integration regression test `pkg/remote/execute_release_test.go` — RED (refs=1) → GREEN (refs=0) proven on nezha.

- **MAJOR (down FAIL-bluff):** `cmdDown` printed "teardown complete" + exit 0 even when a host was unreachable. Fixed — tracks per-host failures, exit 1 when teardown can't be confirmed. Proven: unreachable host → exit 1; reachable → exit 0.
- minors: health-check timeout/cancel now distinguished from a real failure; test 62 free-port probe uses portable `ss -ltn` (not bash-only /dev/tcp). Re-review verdict: **GO — no blocking findings** (all 4 prior fixes confirmed sound, no regressions, white-box test verified not-false-PASS). 2 minor notes also closed: ExecuteStream same-class ref leak on rare cmd.Start failure (Containers 18ed03d) + orchestrator `--port`/`--publish` range-validation. tmux Containers pointer → 18ed03d.

Open follow-ups: none. The binary is on-demand ("when and if we need it").

§3.23 — libtinfo cross-distro static-link fix + test-determinism hardening → v1.0.23 (2026-06-16)

Status: DONE — RELEASED v1.0.23 (2026-06-16). Dual-host re-validated: nezha.local (ALT Linux) synced to v1.0.22 baseline, `setup.sh` run, full destructive suite executed on real Linux hardware.

What landed. (1) **libtinfo fix (A46):** the containerized Linux build linked `libtinfo.so.6` dynamically; on ALT Linux (no NCURSES version nodes) the loader warned on every invocation. Fixed by static terminfo link (`LIBTINFO_LIBS="-l:libtinfo.a"` in `docker/build_inside_container.sh`) — `ldd` shows no libtinfo, jemalloc/glibc stay dynamic, warning gone. New test 61 + verify gate `CM-NO-DYNAMIC-LIBTINFO`. (2) **Test-determinism (A47):** test 27 (macOS cd-poll), tests 12/14 (Linux OOM-poll) — §11.4.1 timing races fixed, now deterministic. (3) macOS test 22 codegraph OpenCode wiring repaired (host `~/.config/opencode/opencode.json`, §11.4.78).

Validation (captured — `docs/qa/2026-06-16-libtinfo-crossdistro/evidence.md`): nezha verify.sh EXIT 0 PASS=49/0/11 (tests 12/14 PASS w/ real kernel OOM-kill, deterministic 5/5); meta-test 52 CAUGHT / 0 ESCAPED; macOS run_all PASS=55/0/5; test 61 PASS; libtinfo warning 0 lines (was 1/invocation).

No open follow-ups. Issues.md zero non-terminal items. Honest SKIPs remain: `08_oom_score_adj` (root/setcap), GUI/clipboard/remote/DB opt-ins.

§3.22 — Apple `container` integration: on-demand containerized Linux under macOS for testing → v1.0.22 (2026-06-13)

Status: DONE — RELEASED v1.0.22 (2026-06-13). Testing-capability addition (no tmux source/wrapper change — shipped binary + `tmx` wrapper unchanged from v1.0.21).

Apple `container` 1.0.0 incorporated into the `vasic-digital/Containers` submodule as a new generic `pkg/crossbuild/apple_container.go` backend + `RunInLinuxContainer` (unit + integration tests + challenge). Proven on this macOS 15.5/arm64 host: real `container run` returns `Linux aarch64` (host is Darwin), host-dir mount round-trips, paired mutation (strip `--mount` → exit 99). Consumed extend-don't-reimplement per §11.4.74 / §11.4.76.

tmx harness `scripts/test_apple_container.sh` builds tmux 3.6a INSIDE an Apple-container Linux VM (genuine Linux ELF: `osdep-linux.o` + `libjemalloc.so.2` + `libevent_core`) and runs the suite. **Evidence** `docs/qa/2026-06-13-apple-container/`: `uname.txt=Linux aarch64`; `tmux-version.txt=tmux 3.6a`; `elf-proof.txt` (ELF magic + jemalloc ldd); `build.log`; `run_all.log` + `summary.txt` = **PASS=30 / FAIL=0 / SKIP=28** (base image `ubuntu:22.04`). 28 SKIPs are honest §11.4.3/§11.4.81 topology gaps (no systemd-cgroup, no physical terminal/clipboard/mouse, no host tooling); the 30 core tmux tests RAN + PASSED. Release notes for GitHub + GitLab at `docs/qa/2026-06-13-apple-container/RELEASE_NOTES.md`. Fixed.md A44; CHANGELOG v1.0.22 (Sources verified 2026-06-13 against apple.github.io/container/ documentation/); VERSION 1.0.22/23.

§3.21 — Copy/paste ARCHITECTURE fix (mouse off default) + HelixCode crash forensics → v1.0.21 (2026-06-13)

Status: DONE — RELEASED v1.0.21 (2026-06-13), committed `b905c9b` pushed to github + gitlab. Full validation captured: setup verify gate PASS=53/FAIL=0/SKIP=5 (`~/.tmux.conf` installed); test 59 wire-level RED→GREEN; test 57 determinism 10/10 clean + 8/8 hostile-clipboard (0 FAIL); collateral sweep 41 PASS/0 FAIL ×2; workable-items DB 14/14 Go tests + byte-identical round-trip; meta sweep 51 CAUGHT / 0 ESCAPED /

8 SKIP (M-MOUSEDEFAULT caught). QA evidence: `docs/qa/2026-06-13-mouse-off-default/`, `docs/qa/2026-06-13-server-death/`, `docs/qa/2026-06-13-helixcode-crash/`. Bug 1 (HelixCode crash) → Issues.md F1 Operator-blocked (operator runs `docs/qa/2026-06-13-helixcode-crash/diagnose.sh`).

Bug 2 (copy/paste) — DONE at source, anti-bluff proven. Operator (2026-05-28 .. 2026-06-13, all emulators): "select multiple lines does not work … must work Linux+macOS … must scroll and always be selectable — right-click → Copy." Systematic-debugging: 3 prior binding-level fixes (v1.0.15/18/20) failed → question the architecture. ROOT CAUSE (wire-level, no guessing): `set -g mouse on` emits mouse-tracking DECSET enables (`CSI ?1000h/?1002h/?1006h`) that SUPPRESS the terminal's native selection + right-click→Copy. FIX: default flipped to `set -g mouse off` (terminal owns the mouse → native select/copy/scroll everywhere); tmux mouse on demand via `prefix m`. PROOF (test 59, real PTY): `mouse on` = 6 enable-DECSET, `mouse off` = 0; after `prefix m` = 3. 4-layer: verify.sh L1 `mouse off` gate + test 59 (RED→GREEN) + meta `M-MOUSEDEFAULT` + regression sweep (56/57/58 enable mouse for on-demand path; 17 + TMUX-CH-17 assert the default; 44/45/47/48 unaffected). Docs (scrolling/clipboard/guide §5.7) updated; Fixed.md A43; CHANGELOG v1.0.21; VERSION 1.0.21/22.

Bug 1 (HelixCode session crashes the whole terminal) — RESOLVED 2026-06-13 (operator-confirmed; Fixed.md A45) — was OPERATOR-BLOCKED (Issues.md F1).** Fresh `tmx new -s HelixCode` creates+attaches cleanly over a real PTY; 5 crash vectors (passthrough/extkeys/attach-reload/rename-format/stale socket) each reproduced headlessly + DISPROVEN as standalone causes. Crash needs operator-side runtime state (the live HelixCode TUI agent). Ready operator diagnostic + forensics: `docs/qa/2026-06-13-helixcode-crash/`.

NEXT (resume here): (1) build/sync workable-items DB; (2) `sync_all_markdown_exports.sh` (docs changed); (3) `bash scripts/setup.sh` (regenerate wrapper for this checkout — current installed wrapper points at a non-existent `/Users/...` path — + reinstall conf); (4) full retest `run_all.sh` + meta on the committed tree (conf mutations need clean tree); (5) `commit_all.sh` push to all upstreams.

§3.20 — tmux PASTE fix + tmx reload + attach-reload → v1.0.20 (2026-05-29)

Status: DONE — RELEASED v1.0.20 (2026-05-29). GitHub: https://github.com/vasic-digital/tmux/releases/tag/v1.0.20 · GitLab: https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.20. Operator-confirmed (drag-select + Cmd-V pastes selected content). Dual-host validated at release commit 0205b44: Mistborn run_all 53/0/4 + meta 50/0; nezha run_all 44/0/13 + test 58 PASS. Root cause of the long copy/paste saga: a tmux server loads config only at startup, so re-opened long-lived sessions kept stale bindings — fixed by `tmx attach` auto-reload + `tmx reload` + plain-drag override + POSIX paste binding. Tests 56/57/58 (real-mouse SGR + clipboard) + meta 50/0.

- **Root cause of "still can't copy"**: the live Herald session was started BEFORE the mouse fix and a tmux server loads config only at start → stale forwarding binding. Fix is reload-of-running-sessions. Added `tmx reload [-t NAME]` (source-file into running sessions; skips dead sockets).
- **PASTE FIX**: `prefix P` binding `tmux load-buffer - <<< "$(#{@clip-read})"` was broken 3 ways — #{@clip-read} nested-quote collision ("completely new value"), `<<<` bash-herestring fails under /bin/sh (dash/Linux), literal `\;`. Rewritten POSIX-clean (single-quoted run-shell wrapping inline double-quoted probe `| tmux load-buffer - && tmux paste-buffer -p`). Real prefix-P now pastes the EXACT clipboard value.
- **Tests**: NEW test 57 (A stale-repro / B reload-fixes-copy / C paste-buffer / D REAL prefix-P exact-value / E no-`<<<` POSIX guard), 3/3. test 46 T4 + verify.sh Layer-1 gate updated to accept the POSIX paste binding + ADDED gates for plain-drag override + prefix+m. Meta mutations M-PASTE + M-PLAINDRAG
  - M-MOUSETOGGLE. Evidence: docs/qa/2026-05-29-paste-fix-reload/.
- **Mistborn**: installed via setup.sh (verify GREEN), run_all 52/0/4, meta 49 CAUGHT / 0 ESCAPED. Copy + paste + reload all proven on the installed host.
- **HONEST NOTE**: during socket cleanup the live `Herald` tmux server ended (no tmux servers running afterward); exact cause undetermined (NOT guessed). Aggressive socket-file pruning around a live session was careless — stopped. Stale Herald socket removed.
- **PENDING (operator gate)**: operator to manually validate copy+paste in a FRESH `tmx new -s X` session; THEN nezha setup + validation; THEN next release.

§3.19 — Dual-host update + full test/verify + v1.0.19 release (2026-05-29)

Status: DONE — RELEASED v1.0.19 (2026-05-29). GitHub: https://github.com/vasic-digital/tmux/releases/tag/v1.0.19 · GitLab: https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.19. §11.4.40 full retest at the release commit `c5235f8`: Mistborn run_all 51/0/4 + meta 48/0; nezha run_all 43/0/12 + meta 48/0; M22 CAUGHT on both. v1.0.19 = the codegraph non-interactive PATH probe in run_all + meta-test (no product/runtime change). Both hosts verified at the release commit. Mistborn (Darwin): run_all `PASS=51 FAIL=0 SKIP=4`, meta `48 CAUGHT / 0 ESCAPED` (M22 CAUGHT). nezha (Linux): run_all `PASS=43 FAIL=0 SKIP=12`, meta `48 CAUGHT / 0 ESCAPED` (M22 CAUGHT). Fix landed this round: added the A31 npm-prefix PATH probe to `scripts/tests/run_all.sh` + `meta_test_false_positive_proof.sh` (commits 5d6196c, a0a3f85) so the codegraph tests (20/22) + M22 mutation resolve the CLI in NON-INTERACTIVE shells too — nezha's `.bashrc` adds `~/.npm-global/bin` only behind an interactive-guard, so standalone run_all over ssh-batch previously failed 20/22 spuriously while real interactive agent usage worked. test 16 nezha full-suite flake ((window-name test, passes standalone)) did not recur this round.

§3.18 — v1.0.18 REAL mouse-copy fix: plain drag selects+copies in Claude Code, proven with a real mouse (2026-05-29)

Status: DONE — RELEASED v1.0.18 (2026-05-29). GitHub: https://github.com/vasic-digital/tmux/releases/tag/v1.0.18 · GitLab: https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.18. Real-mouse PROVEN on BOTH platforms: Mistborn (suite 51/0/4 + meta 48/0 + real cliclick cursor drag over iTerm2 → pbpaste) AND nezha Linux (verify 43/0/12 GREEN + headless SGR real-mouse-drag → @clip sink, after the TERM=xterm-256color test-harness fix). Commits 5c667cc → 06e43d0 → 69d1418. NOTE: test 16 (window-name) flaked once under full-suite load on nezha (passes 3/3 standalone + GREEN on retry) — pre-existing §11.4.50 reliability issue, unrelated to the mouse fix; tracked for a future poll-with-budget fix.

Operator re-reported that select+copy still failed in Claude Code and demanded it work on BOTH Linux + macOS. The v1.0.17 `prefix m` toggle was insufficient (required a magic keystroke). Root cause reproduced with a REAL mouse: in a mouse-tracking app (`mouse_any_flag=1`) tmux's built-in root `MouseDrag1Pane` forwards the drag to the app → no selection. **Fix:** override `bind -n MouseDrag1Pane` `if -F '#{pane_in_mode}' 'send -M' 'copy-mode -M'` in `scripts/tmux.conf.template` — plain drag always enters copy-mode + copies on release; click+wheel still reach the app; no terminal-specific modifier (works Linux + macOS). PROVEN with a real mouse: headless SGR-1006 drag injection (mouse_any_flag=1) → @clip sink, regression-discriminating, 3/3 (test 56, cross-platform); + real `cliclick` cursor drag over iTerm2 → `pbpaste` (macOS GUI layer). Meta-test `M-PLAINDRAG` CAUGHT; meta `48 caught / 0 escaped`; suite `51/0/4`. Commit 5c667cc. Pending: dual-host setup.sh (refresh ~/.tmux.conf), nezha headless test-56 proof, v1.0.18 release.

§3.17 — v1.0.17 two user-reported bug fixes + B3 closure + dual-host re-validation (2026-05-29)

Status: DONE — RELEASED v1.0.17 (2026-05-29). GitHub: https://github.com/vasic-digital/tmux/releases/tag/v1.0.17 · GitLab: https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.17. Dual-host setup GREEN (Mistborn suite 51/0/4 + meta 47/0; nezha 43/0/12, double-prompt real-HOME 2→1). CodeGraph 0.9.7 validate PASS both hosts. Commits 1de20c5 → a6a81df → bbb1467.

Operator reported two product defects 2026-05-29: (1) mouse select/copy unusable in tmx panes, especially under Claude Code; (2) double session-name prompt on new bash-login terminals. Plus directive to validate every fix with autonomous physical-proof tests, run `setup.sh` on Mistborn + nezha, document, and release via gh/glab.

- **A41 (double-prompt) — FIXED.** Root cause (reproduced: nezha `bash -l -i`→PROMPT_COUNT=2): `.bash_profile` carries the tmx-shell-init source line AND sources `.bashrc` which also carries it → one process double-sources; the blank/return path re-prompts. Fix: per-process non-exported `_TMX_SHELL_INIT_PROMPTED` guard in `scripts/tmx-shell-init.sh.template`. Test 54 RED→GREEN (3/3) + meta-test `M-DBLPROMPT`.
- **A42 (mouse copy) — FIXED.** Config was correct & live (not stale); copy pipe works. Cause: in mouse-tracking apps plain drag forwards to the app; iTerm2 `Option Key Sends=Normal` eats Alt-drag → only Shift-drag reached tmux (undiscoverable). Fix: `prefix m` mouse-toggle (native-terminal selection escape hatch) + Shift-drag docs in `scripts/tmux.conf.template`. Test 55 RED→GREEN (3/3) + meta-test `M-MOUSETOGGLE`; test 56 real-mouse honest SKIP (§11.4.3).
- **B3 (P5-M20/M21 escapes) — CLOSED + migrated to Fixed.md §B3.** State-verified `MUTATIONS CAUGHT 45 / ESCAPED 0` on Mistborn

(tests 49/50 close the layer-4 gaps). M22 nezha CodeGraph baseline repaired via `codegraph_setup.sh` during dual-host setup.
- **Mistborn validation:** full suite `PASS=51 FAIL=0 SKIP=4`; meta-test `45 CAUGHT / 0 ESCAPED`. Evidence in `qa-results/` + plan at `docs/superpowers/plans/2026-05-29-tmx-mouse-doubleprompt-b3-release.md`.
- **Pending:** dual-host `setup.sh` (Mistborn redeploy + nezha pull+setup, nezha double-prompt re-prove, M22 repair), CHANGELOG/changelog-doc with both-host evidence, VERSION bump (1.0.17/18), gh+glab release.

§3.1 Governance bring-up — Issues.md / Fixed.md / explicit §11.4.x anchors landed

Status: COMPLETE (2026-05-08T17:06Z).

- Created `Issues.md` (open / in-flight tracker) with:
  - Categories A-E, status conventions OPEN / PARTIAL / BLOCKED / RUNNING / INVESTIGATED
  - Status reclassification rules referencing §11.4.6
  - Seeded items: META-MUT-001, CHAL-COVER-001, TEST-AUDIT-001, TMX-T5, TMX-T7, TMX-T8, TOPO-DISPATCH-001
- Created `Fixed.md` (closed archive) with:
  - A0 initial migration (commit `08d4ba5`, 2026-05-07)
  - B0 §1 covenant propagation (commit `b92bf7f`, 2026-05-08)
  - C0 test 09 crash-isolation-scope landed and green (commit `b92bf7f`, 2026-05-08)
- Updated `Constitution.md`:
  - Explicit §11.4.1, §11.4.2, §11.4.3, §11.4.4, §11.4.5 anchors added (preserving §1 prose verbatim)
  - §11.4.6 no-guessing mandate added with verbatim user-mandate quote
  - §9 absolute data safety added
  - §12.6 60% host memory budget added (TMX_MEM enforcement seam)
  - §5 expanded into full §12.10 continuation-document mandate
- Updated `CLAUDE.md` and `AGENTS.md` to carry every numbered anchor + canonical-authority cross-references

§3.2 Per-session containerization (Phase B — multi-session work)

Status: RESEARCH COMPLETE + WRAPPER LANDED + ISOLATION VERIFIED (2026-05-08).

Web research output: `docs/plans/containerization.md` recommends `systemd-run --user --scope` (cgroup-v2 transient scope) over podman-per-session. Wrapper at `scripts/tmx` implements `tmx {new|attach|ls|kill}` with `MemoryMax=$TMX_MEM` (default 8G), `CPUQuota=$TMX_CPU` (default 200%), `TasksMax=4096`, `Delegate=yes`.

Test 09 (`scripts/tests/09_crash_isolation_scope.sh`) — 14 PASS / 0 FAIL / 0 SKIP — verifies T1 (host capability), T2 (wrapper invariants), T3 (cgroup interface evidence), T4 (SIGKILL containment + user.slice survival), T6 (concurrent registration independence). Source-line breakdown:

- **T1:** systemd 258 + cgroup v2 mounted ✓
- **T2:** tmx wrapper invokes `systemd-run --user --scope` + sets MemoryMax/CPUQuota/TasksMax/Delegate=yes ✓
- **T3:** transient scope created; `/sys/fs/cgroup/.../memory.max` reads 268435456 bytes (matches set 256M) + `/sys/fs/cgroup/.../cpu.max` reads `50000 100000` (50% quota over 100ms period) — both POSITIVE EVIDENCE per §1 ✓
- **T4:** spawn scope, read MainPID from `cgroup.procs`, SIGKILL it, verify scope inactive AFTER kill, verify `default.target=active` THROUGHOUT (user.slice survives — Constitution §1 invariant) ✓
- **T6:** 3 concurrent scopes, all registered + active simultaneously ✓

§3.3 Pending tests — moved to Issues.md

The previously-listed deferred tests (T5 memory pressure under cap, TasksMax stress, Concurrent OOM independence) have migrated to `Issues.md`:

- **TMX-T5** — Issues.md C1 (PARTIAL; operator-unblock runbook for dedicated test host)
- **TMX-T7** — Issues.md C2 (OPEN; new test 11 planned)
- **TMX-T8** — Issues.md C3 (OPEN; new test 12 planned)

Cross-cutting blockers also tracked in `Issues.md`:

- **META-MUT-001** — Issues.md A1 (paired-mutation harness needed before §11.4.4 layer-4 coverage is real)

- **CHAL-COVER-001** — Issues.md B1 (HelixQA Challenge entries pending)
- **TOPO-DISPATCH-001** — Issues.md D1 (formal topology dispatch matrix)

§3.4 Anti-bluff enforcement — test audit + challenges fix + covenant propagation verification

Status: COMPLETE (2026-05-08T18:00Z).

- Verified anti-bluff covenant propagation across ALL governance files:
  - Root `Constitution.md`: §1 + §11.4.1-§11.4.6 present ✓
  - Root `CLAUDE.md`: full covenant present ✓
  - Root `AGENTS.md`: compact covenant present ✓
  - `Containers/CLAUDE.md`: full covenant present ✓
  - `Containers/AGENTS.md`: full covenant present ✓
  - `tmux/` (upstream submodule pinned to `3.6a`): N/A — no governance files, not modified per Constitution
- Performed line-by-line §11.4.2 anti-bluff audit of tests 01-09:
  - All 9 tests confirmed to carry positive runtime evidence (/proc files, cgroup interfaces, tmux command output, systemctl state).
  - No FAIL-bluff vectors found (all `set -uo pipefail`, guarded variables).
  - Full audit table moved to `Fixed.md` B2.
- Fixed broken paths in `scripts/challenges/tmux.yaml` (`scripts/tmux/tests/` → `scripts/tests/`).
- Migrated TEST-AUDIT-001 from `Issues.md` B2 → `Fixed.md` B2.

§3.5 Hostname-derived status-bar colour — algorithm + wrapper + 2 tests + challenges

Status: COMPLETE (2026-05-08T19:00Z).

- Created `scripts/hostname_color.sh` — DJB2 hash over hostname → curated 27-colour palette. Deterministic, testable standalone.
- Updated `scripts/tmx.template` — added `_apply_host_color()` function that invokes `hostname_color.sh` and applies `set -g status-style bg=<colour>` on the running tmux server. Applied on both new-session and attach paths.
- Created `scripts/tests/10_hostname_color_algorithm.sh` — 5 invariants (deterministic, valid `colourNNN`, palette member, spread ≥12/16, empty-fallback to system hostname). All 5 PASS on this host.

- Created `scripts/tests/11_hostname_color_integration.sh` — verifies wrapper applies correct colour to tmux server. SKIPs if binary/wrapper not yet built.
- Updated `scripts/tests/run_all.sh` — glob changed from `0[1-9]_*.sh` to `[0-9][0-9]_*.sh` to support tests 10+.
- Updated `scripts/challenges/tmux.yaml` — added TMUX-CH-10 (algorithm) and TMUX-CH-11 (integration).
- Updated `docs/research/customization/colors.md` — added implementation documentation covering algorithm, integration, verification, usage, and anti-bluff covenant.
- Updated `AGENTS.md` — bump test count from 9 to 11.

§3.6 Full coverage — missing challenge entries, destructive tests, topology dispatch

Status: COMPLETE (2026-05-08T20:00Z).

- **CH-09**: Added missing challenge entry for crash isolation scope (test 09) — was the only test without a challenge.
- **Test 12** (`12_memory_pressure_under_cap.sh`): TMX-T5 — allocates up to MemoryMax+10% inside a transient scope, captures dmesg OOM-kill evidence, validates user.slice survival. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **Test 13** (`13_tasksmax_stress.sh`): TMX-T7 — fork-bomb resistance. Spawns processes up to TasksMax=4096, reads pids.current/pids.max from cgroup interface. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **Test 14** (`14_concurrent_oom_independence.sh`): TMX-T8 — three scopes A/B/C, OOM-kills A, verifies B and C survive with original MainPIDs. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **CH-12/13/14**: Challenge entries for all three destructive tests.
- **Topology dispatch**: added `_probe_topology()` to `scripts/tmx.template` — detects systemd version + cgroup v2, classifies host as `tmx-supported/tmx-degraded/tmx-unsupported` per §11.4.3.
- **Fixed.md B2**: closure SHA updated to `68a65b0`.
- **Issues.md → Fixed.md migration**: B1 (CHAL-COVER-001), C1 (TMX-T5), C2 (TMX-T7), C3 (TMX-T8), D1 (TOPO-DISPATCH-001) moved to Fixed.md.

§3.8 Audit cycle 2026-05-13 — submodule pin-drift caught + governance staleness fixed

Status: COMPLETE (2026-05-13T00:00Z).

Triggering event: user invoked `/init` followed by "what is left unfinished, no-bluff policy heavily enforced everywhere". Audit found and remediated:

- **CRITICAL §1 / §11.4.6 bluff**: `f4132aa Auto-commit` (2026-05-13) silently bumped tmux submodule from `cc117b5` (tag `3.6a`) to `3f651d9f` (`3.6a-329-g3f651d9f` — 329 commits past the only stable tag). Governance docs across Constitution / CLAUDE / AGENTS / CONTINUATION / README still claimed "pinned to tag 3.6a". User decision: revert to tag (newer stable tag does not exist; 329 commits are unreleased upstream master). Submodule pointer reverted to `cc117b5`.
- **README.md staleness**: line 5 ("eight verification tests"), line 33 (`PASS=6 FAIL=0 SKIP=2`), line 37 ("8 tests cover"), line 39 (2-SKIP list) — all rewritten to current 14-test / PASS=10 SKIP=4 / 4-SKIP-list state with explicit `TMX_TEST_DESTRUCTIVE=1` callout and §11.4.4 layer-4 harness reference.
- **CLAUDE.md drift**: added top-of-file project summary + canonical-doc cross-links + fresh-conversation workflow; fixed `0N_*.sh` glob → `NN_*.sh`; named the four layers inline in the Test-interrupt rule; added "Files to never edit directly" section (parity with AGENTS.md).
- **AGENTS.md drift**: mirrored CLAUDE.md project summary + cross-links; fixed same `0N_*.sh` glob bug; added meta-test row to commands table.
- **Issues.md format**: restored A / B / C / D / E section headers as empty sections (was: A / B / E only, with C and D missing despite conventions list referencing them). All bodies note "landed in `Fixed.md`" with item IDs for traceability.
- **CONTINUATION.md timestamp**: 2026-05-08T22:00Z → 2026-05-13T00:00Z; this entry added under §3.8 per §12.10 invariant.
- **M6 mutation added** (`scripts/tests/meta_test_false_positive_proof.sh`): injects `$$` into the hostname_color hash, making the algorithm non-deterministic per-invocation. Test 10 T1 must FAIL under this mutation. Closes the audit-flagged gap: until M6, the "same-host = same-color" user invariant was protected by side-effect only (no dedicated anti-randomness mutation).
- **`docs/guide/README.md` phantom-script bluff fixed**: 3 occurrences each of `verify_tmux.sh` → `verify.sh`, `setup_tmux.sh` → `setup.sh`, `install_tmux_deps.sh` → `install_deps.sh`; "8 tests" → "14 tests" (2×).
- **GUIDE.md "severity hierarchy" pre-existing bluff caught** (Fixed.md A3): documented "blockers / critical / advisory"

classification does not exist in `run_all.sh` — every test is treated equally (any FAIL = RED). Rewritten to describe the actual gate logic. Caught while extending the table for tests 09-14; would have been silently doubled otherwise.

- **Verification + validation cycle 2026-05-13** (Fixed.md A14): operator invoked `superpowers:verification-before-completion` + asked for full anti-bluff verification + governance propagation. Every verification command run FRESH this session per Iron Law. Captured evidence: `setup.sh --verify-only` SUMMARY=PASS=10 FAIL=0 SKIP=5 GREEN; `test_e2e.sh` PASS=9/0/0 GREEN; test 15 PASS=6/0/0; live `tmx new -s VerifyDemo` shows operator's host shell (`milosvasic@Mistborn`, `which brew=/opt/homebrew/bin/brew`, `ulimit -t=86400`, `ulimit -u=2666`). Two defects caught and fixed: (D1) wrapper hostname resolution drifted to FQDN after bridge removal — restored scutil-LocalHostName path so colour stays `colour202` for Mistborn; (D2) `install_deps.sh` + `build_oom_set.sh` weren't OS-aware out of the box — now `install_deps.sh` detects Darwin → brew install (no sudo), `build_oom_set.sh` SKIPs on non-Linux. Governance propagation completed: verbatim user-mandate quote now in root CLAUDE.md + AGENTS.md (previously only in Constitution + Containers); §11.4.7 reference added to Containers/CLAUDE.md + Containers/AGENTS.md (previously only Containers/CONSTITUTION.md). Install state: shell snippets in both ~/.bashrc + ~/.zshrc, `zsh -c 'which tmx; tmx -V'` returns native wrapper + `tmux 3.6a`. macOS RLIMIT_AS gap documented honestly per §1 (XNU EINVAL for unprivileged setrlimit on memory).
- **Per-session isolation + Constitution §11.4.7** (Fixed.md A13): user reported `core@localhost` + lack of OOM containment. Forensic: `tmx new -s isol1 -d` + `tmx new -s isol2 -d` previously shared ONE cgroup (`run-p504653-i504654.scope`) — README's "if one session OOMs, others survive" was a §1 bluff. Architectural rewrite: each `tmx new -s NAME` now spawns its OWN tmux server on socket `tmx-NAME` inside its OWN scope `tmx-NAME.scope` with host-adaptive `MemoryMax` (max(MemTotal × 60% / 4, 2GB)), `CPUQuota=200%`, `TasksMax=4096`, `Delegate=yes`. OOM in any session = contained to that scope; B+C survive with original MainPIDs (verified test 14 PASS=8/0/0 with stress-ng `tmx send-keys`). Test 15 NEW: per-session cgroup distinctness (6 assertions, positive readback from /sys/fs/cgroup). Test 11 + test 14 + test 08 rewritten to OPERATOR PATH (`tmx new -s NAME` instead of hand-crafted `systemd-run --user --scope`). e2e T7 NEW: bridge confirms two distinct scopes via VM systemctl. Meta-test M5 retargeted, M7 retargeted, M9+M10 NEW (10/10 mutations caught). Plan + operator decisions captured in `docs/plans/per-session-isolation.md`. **Constitution §11.4.7 NEW**

(operator-path test coverage rule): every gate test MUST exercise the same entry point an end-user invokes; tests that hand-craft equivalents are supplementary. Propagated to `Containers/CONSTITUTION.md` at same depth. Forensic anchor: Fixed.md A12 (status-bar green default) + A13 (sessions sharing one cgroup) — both shipped while non-operator-path tests reported GREEN.

- **Regression protection for A10 — gate-coverage gap closed** (Fixed.md A11): user demanded "make sure nothing passes again! zero-bluff policy MUST BE followed blindly!" — honest accounting of why the existing gates missed the colour bug:
  - test 11 always passed `-S "$SOCKET"` → tested explicit-socket path only; operator's default-socket use case uncovered.
  - meta-test M4/M5 targeted `scripts/tmx` (the SSH bridge on Darwin) instead of `scripts/tmx-vm` (the actual VM wrapper) — paired-mutation harness was mutating the wrong file.
  - test_e2e.sh pre-fix didn't read `status-style` at all.
  - Triple-layer regression protection added: (1) test 11 T6 — new test exercising default-socket path with positive evidence (`tmux show -g status-style` with NO `-S`, FAILs explicitly on `bg=green` default-applied bluff); (2) meta-test M4/M5 retargeted to `scripts/tmx-vm`; (3) M7 (re-introduce SOCK-empty early return) + M8 (hardcode bg=green) both caught by test 11 T4.1/T6. Result: all 8 mutations caught + reverted (16 PASS), full destructive suite PASS=14/0/0, e2e PASS=8/0/0.
- **Status-bar colour silently green (default) + bridge ignored macOS hostname** (Fixed.md A10): user said "Bottom was green for current host. Is that expected for mistborn?" — two stacked bluffs caught:
  - `_apply_host_color` / `_apply_oom_score` had `[ -n "$sock" ] || return 0` — silently bailed when `tmx new -s Test` was invoked WITHOUT `-S` (which is the primary operator use case; tmux uses its default socket then). So `set -g status-style` was never fired → tmux default `bg=green` shown. Test 11 always passes `-S "$SOCKET"` so the bug was uncovered. Fix: helpers now build `target=(-S "$sock")` only when sock is non-empty; otherwise omit `-S` and let tmux use its default socket.
  - Bridge `scripts/tmx-mac.template` didn't forward macOS hostname → `hostname_color.sh` inside VM resolved against `localhost.localdomain` regardless of macOS host. Every macOS user got the same VM-derived colour, violating host-distinguishability. Fix: bridge captures `scutil --get LocalHostName` (e.g., "Mistborn"), forwards as `TMX_HOSTNAME=<host>` in the remote SSH command. Wrapper's `_apply_host_color` passes `${TMX_HOSTNAME:+"$TMX_HOSTNAME"}` to

hostname_color.sh — env var is source of truth when set, `$(hostname)` fallback otherwise.
    - New `test_e2e.sh` T4.5 explicitly reads `tmx show -g status-style`, computes expected colour from `scutil --get LocalHostName` (or `hostname` on Linux), FAILs on `bg=green` (default-applied bluff) or non-match. Captured runtime evidence on Mistborn: `status-bg 'colour202' matches hostname-derived 'colour202' for 'Mistborn'`.
- **Interactive `tmx new` fixed + e2e automation harness** (Fixed.md A9):
    - User ran `tmx new -s Test` interactively after install and hit `open terminal failed: not a terminal`. The bridge + ssh -t allocated TTY correctly; the WRAPPER's `cmd & wait` pattern was disconnecting tmux from the foreground process group → no TTY access. Detached-mode tests (`-d`) all passed because tmux daemonizes; interactive (no `-d`) hit the bluff. §11.4.1 FAIL-bluff in the wrapper itself.
    - Fix: wrapper now detects `-d`/`-D` in args (INTERACTIVE flag), and for interactive: runs systemd-run in foreground inheriting TTY, injects `-d` after the new-session keyword (so tmux server detaches), then `exec attach` to take over TTY interactively. For detached: keeps existing path.
    - New: `scripts/test_e2e.sh` — 7-test operator-facing automation: prerequisites + `tmx -V` (via bridge) + new-session + send-keys + capture-pane + survive-without-client + kill-session. All with positive runtime evidence; trap-on-EXIT cleanup. Captured the literal marker string echoed inside the session pane through the bridge — proves the full operator stack carries intent end-to-end.
    - Verified: `bash scripts/test_e2e.sh` GREEN 7/0/0; `TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh` still GREEN 14/0/0 (wrapper change didn't regress existing tests).
- **macOS `tmx` fully operational + side-by-side end-to-end** (Fixed.md A8):
    - `bash scripts/setup.sh` on Darwin now diverges intelligently: builds (containerized), generates BOTH `scripts/tmx-vm` (Linux wrapper for VM paths) AND `scripts/tmx` (Darwin SSH bridge from new `scripts/tmx-mac.template`); runs `scripts/test_vm.sh` for §11.4 target-env verification (since binary cannot exec on Darwin); on GREEN, appends bashrc snippet to BOTH `~/.bashrc` AND `~/.zshrc` (macOS defaults to zsh).
    - **Bridge mechanics**: `scripts/tmx-mac.template` discovers podman machine's SSH endpoint via `podman machine inspect` at every call (port re-discovery — survives

`podman machine stop && start`), verifies `tmx-vm` is executable in the VM, then `ssh -t -i <identity> -p <port> core@127.0.0.1 "<vm-repo>/scripts/tmx-vm <quoted-args>"`.
    - **Verified end-to-end on Darwin 24.5.0 / Apple Silicon**: `zsh -c 'source ~/.zshrc; which tmx; tmx -V'` → `/Users/.../scripts/tmx` + `tmux 3.6a`; system `/opt/homebrew/bin/tmux -V` → `tmux 3.6a` — side-by-side coexistence proven; VM-side verify GREEN PASS=11 FAIL=0 SKIP=3.
    - **`scripts/verify.sh` now respects `$WRAPPER` and `$TMUX_BIN` env vars** so test_vm.sh / bridge tooling can redirect to the right wrapper without source modifications.
    - **README.md** gained a new "Architecture" section with ASCII diagram showing Linux-native vs Darwin-bridge dispatch; `docs/guide/README.md` §5.5 documents the bridge mechanics + Mermaid flowchart; CLAUDE/AGENTS one-liners updated.
- **Final sweep — env-specific wrapper + §255 violations + sed portability** (Fixed.md A7):
    - **Environment-bound wrapper:** `scripts/tmx` was generated for one env (container `/repo` paths OR VM `/Users/.../` paths) and silently failed in the other. New `scripts/test_vm.sh` orchestrator regenerates the wrapper with VM-native paths before each VM-side run. Both `test_containerized.sh` and `test_vm.sh` now own the wrapper-regeneration step for their respective target.
    - **Constitution §255 violations:** "ATMOSphere" in GUIDE.md title + body, challenge yaml description, oom_set.c license comment, tmux.conf.template attribution; `atm_tmux_test_*` / `atm_test_*` / `atm_tmx_test_color_*` socket/session names across 8 test files; `~/.tmux.conf.pre-atmosphere` backup naming. All renamed: code uses `tmx_*` / `pre-vasic-digital`; docs branded "vasic-digital".
    - **`sed -i` portability:** setup.sh used GNU-only `sed -i` for bashrc markers — would fail on macOS BSD sed. Centralized in `_strip_bashrc_snippet()` using `perl -i -ne ... ../ ...` flip-flop range delete (portable GNU+BSD).
    - **Verified post-fix in VM:** `TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh` → PASS=14 FAIL=0 SKIP=0; `META=1 bash scripts/test_vm.sh` → 12/12 mutations caught.
- **Install-mechanism side-by-side bluffs** (Fixed.md A6): user asked about the alias and side-by-side coexistence requirement. Audit found three real defects:
    - bashrc snippet pointed PATH at `scripts/tmux/` (phantom subdir; wrapper is at `scripts/tmx`).

    - `alias tmux='tmx'` line shadowed the system tmux command → broke side-by-side.
    - setup.sh closing message said "Then 'tmux' will use the ATMOSphere build" (wrong command name + stale branding).
    - **Canonical answer to user's question**: the alias is **`tmx`** (not `tmux`). Wrapper at `scripts/tmx` generated from `tmx.template`. `tmx new|attach|ls|kill` invokes the verified vasic-digital tmux; `tmux` invokes whatever was on the operator's PATH before (system / Homebrew / apt / dnf). Both coexist.
- **Full destructive + meta-test cycle caught 6 more FAIL-bluffs** (Fixed.md A5): pushing past PASS=10 SKIP=4 into actually running TMX_TEST_DESTRUCTIVE=1 + meta-test in the podman machine VM (Fedora CoreOS 42 + systemd 257 + stress-ng + tmx-oom-set setcap) surfaced multiple §11.4.1 FAIL-bluffs. Final: **PASS=14 FAIL=0 SKIP=0** (full verify) + **12/12 mutations caught** (meta-test). Real defects fixed:
    - Tests 12/14 `_skip;<continue>` bug — printed SKIP but didn't exit; continued running and FAILed on missing stress-ng → false §11.4.1 FAIL-bluff. Fix: explicit `exit 0`.
    - Tests 12/14 unprivileged dmesg → fallback to journalctl -k via `_kring_count` / `_kring_tail` helpers; broader OOM regex.
    - Test 13 fork-storm OOM-killed its own scope (4096 sleeps × 700KB > 256M MemoryMax); two-phase scope, polling registration, test TASKS_MAX 4096→256 with MemoryMax=512M (production wrapper still TasksMax=4096 — separately grep-verified by test 09 T2.2).
    - Meta-test `sed -i` strip-exec-bit → orig_mode capture/restore via stat + chmod.
    - Meta-test M5 sed pattern was eval-expanded (`${TMX_MEM:-8G}` → `8G`) so it never matched → silent no-op → mutation escaped (a **bluff in the gate itself**). Fix: simplified to literal `MemoryMax=|MemMax=` pattern.
    - Meta-test had no debug visibility → added opt-in `TMX_META_DEBUG=1`.
    - **The user's "same-host = same-color" invariant is now PROVEN end-to-end** with `tmux show -g status-style` reading `colour166` on first session AND on second session on same host (test 11 T4.1 + T5).
- **Build pipeline bluffs caught while reproducing on macOS** (Fixed.md A4): three real defects surfaced when user asked why we couldn't build on macOS:
    - `docker/Dockerfile`: `groupadd -g 20` collided with Ubuntu's `dialout` group → build aborted at step 7/10. Fix: `-o` (non-unique)

on groupadd + useradd. Build now reproducible on Darwin hosts.
    - `docker/build_inside_container.sh`: LDFLAGS `-ljemalloc` was being dropped by linker's default `--as-needed` because tmux doesn't reference jemalloc symbols. **README's "Build-time -ljemalloc" claim was a §1 bluff until this commit.** Fix: `-Wl,--no-as-needed -ljemalloc -Wl,--as-needed`. Verified `ldd` now shows `libjemalloc.so.2` in DT_NEEDED.
    - `docker/build_inside_container.sh`: `make -j2` silently no-op'd after LDFLAGS change (existing binary defeated mtime check). Fix: prepend `make clean` so LDFLAGS changes always take effect.

The `f4132aa Auto-commit` itself is a §12.10 / §11.4.6 violation — opaque commit message, silent submodule mutation, no CONTINUATION update. Cannot rewrite history per §9 data safety; documenting here is the audit trail.

§3.7 §11.4.4 layer-4 paired-mutation harness landed (A1 META-MUT-001)

Status: COMPLETE (2026-05-08T21:00Z).

- Created `scripts/tests/meta_test_false_positive_proof.sh` — §11.4.4 layer-4 harness with 5 registered mutations:
    - **M1**: break hostname_color output format → test 10 T2 FAILs (invalid format)
    - **M2**: force hash to zero → test 10 T3 FAILs (zero spread)
    - **M3**: single-entry palette → test 10 T3 FAILs (palette mismatch)
    - **M4**: remove systemd-run flag from tmx.template → test 09 T2 FAILs
    - **M5**: remove Delegate=yes from tmx.template → test 09 T2 FAILs
- Results on this host: **10 PASS / 0 FAIL / 0 SKIP** — all mutations caught.
- Updated `CLAUDE.md`: removed "PENDING" references, wired mandatory operations to the actual file.
- Updated `AGENTS.md`: layer 4 now marked as landed, not PENDING.
- Updated `Fixed.md`: A1 entry with closure details; all "PENDING META-MUT-001" references replaced with actual coverage references.
- Updated `Issues.md`: A1 removed (→ Fixed.md A1). Issues.md now has **zero open items** — all originally seeded items are closed.
- Cleaned up `CONTINUATION.md` §3.6: removed stale "remaining open" note.

§3.16 — v1.0.15 multi-line + paste-IN + alt-screen Claude Code + workable-items SQLite + DOCX + constitution sync (2026-05-28)

Status: IN PROGRESS — Mistborn build GREEN + Mistborn full suite GREEN + meta-test GREEN; nezha deploy pending; release tag pending.

Triggering event: operator mandate (verbatim, 2026-05-28):

> "In last iteration of work we did we have allegedly fixed copy / paste from and to the tmux session window. Selecting multiple lines and copying of them does not work. We MUST BE able to scroll vertically everywhere and copy / paste anything! Especially in Claude Code (claude command)! Do in depth investigation and systematic debugging, reproduce issue and then apply comprehensive fixes! ... Every bug / issue we report MUST BE first covered with all supported tests which will actually confirm failure / problems we report! Then, after we apply fixes, the same tests MUST CONFIRM all changes and fixes really working! We MUST for every issue or change, for any workable item create proper workable item in our Issues doc and all relevant docs around it (with them all being exported in all expected file types - PDF, HTML, DOCX)! We MUST NEVER forget the flow: workable item → SQLite database → all docs we have related to workable items!"

Five-PWU pipeline executed per §11.4.58 + §11.4.94 (parallel-by- default + zero-idle). Phase 0 fetched + ff-merged `constitution/` from `84c948d` → `6828ff2` (19 commits) and `Containers/` from `fbef9d6` → `2e9ca0e` (17 commits). PWU-A (tmux conf + 4 tests + 4-layer coverage) + PWU-B (§11.4.87..98 propagation) + PWU-C (Go binary FULL Phase 3+ project-local) + PWU-D (DOCX export) ran in parallel; PWU-E (commit + deploy) is the closing sequential PWU.

- **A37 — Multi-line + PASTE-INTO + alt-screen TUI mouse-drag override** (Fixed.md A37): the user's primary ask. Identified the three uncovered surfaces vs v1.0.14: (1) multi-line drag in Claude-Code-like TUIs (mouse_any_flag=1 forwards drag to app), (2) paste-INTO from OS clipboard (set-clipboard external is copy-OUT only), (3) the alt-screen + mouse-tracking surface never asserted end-to-end. Added M-/S-MouseDrag1Pane overrides + matching drag- end @clip-routed binds + @clip-read user-option + prefix+P paste bind. Tests 45/46/47/48 land 4-layer coverage with pbpaste-back physical evidence: PASS=6/0/0 + 6/0/0 + 8/0/0 + 9/0/0 = 29 PASS, RED→GREEN proven for tests 46 + 48. Synthetic alt-screen surrogate `synthetic_alt_screen_app.py` substitutes for Claude Code in tests (no OAuth flake per §11.4.98). Mutations M46 + M48 CAUGHT + RESTORED.

- **A38 — Constitution + Containers sync + §11.4.87..98 propagation** (Fixed.md A38): submodule pointers advanced via `--ff-only`; short-form mirrors for 12 new universal anchors landed in project Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md (each anchor appearing ≥3 times in each consumer per PWU-B audit). New Layer-1 gate `L1C` in verify.sh enforces literal presence.
- **A39 — Workable-items SQLite SSoT (Go binary, project-local Phase 3+)** (Fixed.md A39): constitution scaffold ships Phase-2 stubs only; project added `cmd/workable-items/` with full Phase 3+ logic (sync md-to-db / sync db-to-md / diff / validate / add / close / report). 11 Go sources + embedded schema (drift-check-cited from constitution `6828ff2`) + 10 tests passing 30/30 (3 iters per §11.4.50). Pure-Go `modernc.org/sqlite` (no CGO). Initial DB at `docs/workable_items.db` seeded with 45 items (TMX-001..TMX-045) from live Issues.md + Fixed.md. TRACKED in git per §11.4.95. Honest gaps logged: (a) legacy items default to Type=Task (no Type lines pre-§11.4.16); (b) live-corpus round-trip not byte- identical for free-form bodies (per §11.4.93 phase 6); (c) upstream PR + integration into commit_all.sh deferred.
- **A40 — DOCX export** (Fixed.md A40): `scripts/sync_all_markdown_- exports.sh` extended to emit `.docx` siblings alongside `.html` + `.pdf` via pandoc. 44/44 candidates produced valid DOCX. Layer-1 gate `CM-DOCX-EXPORT-SYNC` enforces canonical-doc sibling presence.

Multi-host deploy + release plan (PWU-E, sequential):

1. ✓ Mistborn `setup.sh --rebuild` GREEN.
2. ✓ Mistborn full suite via `verify.sh` — **PASS=45 FAIL=0 SKIP=3** (3 honest topology SKIPs: oom_score_adj Linux-only, destructive-gated, nezha-required).
3. ✓ Mistborn meta-test — 43 CAUGHT / 2 ESCAPED (pre-existing P5-M20+P5-M21 from v1.0.9) / 8 SKIPPED.
4. ✓ Mistborn `go test ./cmd/workable-items/... -count=3` — 30 PASS / 0 FAIL.
5. commit_all.sh push (this commit).
6. Nezha SSH (`~/nezha` script) → `cd ~/Projects/tmux && git fetch --all && git pull && bash scripts/setup.sh --rebuild && bash scripts/setup.sh --verify-only` — full gate capture expected. T5 honestly SKIPs on headless Linux per §11.4.3 — T2/T3/T4 still PROVE the binding chain.

7. Release tag `v1.0.15` via GitHub + GitLab CLIs with comprehensive `docs/changelogs/v1.0.15.md` referenced.
8. Post-release version bump (per operator mandate "After releasing... increase version (version code) and commit and push all work").

§3.15 — v1.0.14 clipboard copy-OUT physical proof + e2e stale-prereq fix + multi-host deploy (2026-05-22)

Status: COMPLETE (2026-05-22).

Triggering event: operator mandate (verbatim, 2026-05-22):

> "We can always copy / paste from and to the terminal window and current tmux (tmx) session! Using mouse or keyboard MUST WORK properly!!! Scrolling the content / history MUST NOT be broken or anything else! After all changes are done, covered with full validation and verification tests make sure we setup new version to current host (macOS) and nezha (nezha.local is back online). Once both hosts are updated with the latest versions and on both hosts all tests are executed with success with no failure and bringing real proofs of everything working as requested and expected (physical proofs) with no bluffs commit and push all Submodules and main repo to all upstreams and release new version with comprehensive in-depth version (change) log using GitHub and GitLab CLIs."

- **A35 clipboard copy-OUT physical proof** (Fixed.md A35): identified the §101 PASS-bluff hole — bindings existed and grepped fine, but no test ever read back the OS clipboard.
  - NEW `scripts/tests/44_clipboard_copy_out_physical.sh` — operator-path. Spawns `tmx new -s NAME -d`, prints a unique marker, enters copy-mode, search-backward + select-line, invokes `@clip` (T3 direct -X + T4 the literal `y` keystroke that triggers the bind table end-to-end), then reads the OS-native paste tool (T5 — `pbpaste` / `wl-paste -n` / `xclip -o` / `termux-clipboard-get`) and asserts the marker is there. T5 SKIPs honestly with reason on headless Linux; T3/T4 binding- chain proof runs everywhere so the test is never inert. Pre-test save + post-test restore of the operator's clipboard.
  - PASS=7/0/0 this cycle (Darwin); T5 returned the marker via `pbpaste` — physical end-user evidence.
  - Layer-1 verify.sh static gate extended (4 new `_l1` checks for `@clip` + the three bindings).
  - Layer-3: `TMUX-CH-44` in `scripts/challenges/tmux.yaml`.

    - Layer-4: `M44` in `meta_test_false_positive_proof.sh` strips the `@clip` definition line — test 44 T1 catches universally (structural grep), T5 additionally catches where a clipboard tool is reachable. CAUGHT + RESTORED both directions.
- **A36 e2e stale podman-machine prerequisite** (Fixed.md A36): `scripts/test_e2e.sh` T1.2 still hard-required a running podman machine, a leftover from the pre-v1.0.7 SSH-bridge architecture. Fixed by probing for the native Mach-O binary FIRST and only falling back to the podman check for bridge-era installs. e2e GREEN after the fix.
- **B3 P5 escape transparency** (Issues.md B3): meta-test reported 2 ESCAPES on Mistborn — P5-M20 (non-TTY guard) and P5-M21 (cwd-capture hook). Nezha additionally reports M22 as an ENVIRONMENTAL escape (CodeGraph state). All three pre-exist this cycle. Investigation: P5-M20/M21 are layer-4 test-DESIGN gaps — the assertions cannot distinguish "guard fired" from "fallback fired", so a single-layer strip slips through. M22 is an environmental state-dependency on the CodeGraph index/config on nezha. Underlying features GREEN. Surfaced transparently in `Issues.md` B3 + the v1.0.14 CHANGELOG — anti-bluff disclosure per §11.4 / §101 / §11.4.6. Closure conditions documented; deferred to a future cycle.
- **Multi-host deploy** (this turn's operator ask, satisfied): Mistborn updated to v1.0.14 via `setup.sh --rebuild`. Nezha fetched gitlab/main, fast-forwarded to v1.0.14, rebuilt and installed. Both hosts run the same code; both gates GREEN; both e2e GREEN; clipboard physically proven on Mistborn (pbpaste) and binding-chain proven on nezha (tmux buffer, T5 honest-SKIP on headless).

§3.14 — v1.0.8 uniform tmux UI recolouring (active border + clock + window-status-current) (2026-05-21)

Status: COMPLETE (2026-05-21T21:00Z).

Operator-driven follow-up to v1.0.7: "flying animated top decoration"

- clarification "Do coloring of all UI tmux parts with proper color we use instead of default green. Anything colored with that green colors has to become the color we have assigned to the bottom view we are coloring."

Pre-v1.0.8 `_apply_host_color()` set only `status-style bg=$color`. Other default-green tmux surfaces (active pane border, clock face, selected-window highlight) stayed green. v1.0.8 extends the wrapper to apply the hostname-derived colour atomically to all four:

- `status-style              bg=$color`
- `pane-active-border-style  fg=$color`
- `clock-mode-colour         $color`
- `window-status-current-style bg=$color,fg=black`

`mode-style` + `message-style` deliberately untouched (default yellow, not green; yellow provides best contrast against any palette bg).

NEW test 26 (`26_ui_color_uniformity.sh`) — operator-path session spawn + live `show -gv` readback per surface — PASS=5/0/0 on Darwin. NEW M24 paired mutation — regex-strips the three v1.0.8 set-lines from the generated wrapper, asserts test 26 T2/T3/T4 FAILs. MUTATION CAUGHT + FEATURE INTACT both directions.

Captured operator-path readback (Mistborn, this session):

- status-style: `bg=colour44`
- pane-active-border-style: `fg=colour44`
- clock-mode-colour: `colour44`
- window-status-current-style: `bg=colour44,fg=black` → All four surfaces uniformly turquoise (colour44 = RGB 0,215,215).

Verification (Darwin quintuple-fresh, 2026-05-21):

- setup.sh --verify-only → PASS=24 FAIL=0 SKIP=2
- meta-test → 36 caught / 0 escaped / 6 skipped
- e2e → PASS=9 FAIL=0 SKIP=0
- codegraph_validate → PASS=4 FAIL=0 SKIP=1

Released v1.0.8; setup.sh re-run as final install per operator "Make sure we setup the new version once all done." directive.

§3.13 — v1.0.7 cross-host portability + hostname-colour rebalance + Node-22 pin + release (2026-05-21)

Status: COMPLETE (2026-05-21T18:30Z).

Operator-driven cycle: tried to install on Nezha (Linux), 3 tests FAILed → 6 fix rounds → Nezha GREEN → operator reported orange-color collision (Nezha + Mistborn) → palette rebalanced + test 25 + M23 mutation → release v1.0.7.

Six fix rounds (all Nezha-driven; live forensic capture):

1. Initial 3-fix attempt (test 09 timing, test 17 ingestion race, test 21 codegraph bootstrap). Test 17 fixed; tests 09 + 21 had secondary defects.
2. Bash `-lt` fractional bug in round-1 poll-loop (silent fail); npm-prefix PATH augmentation for non-interactive shells.
3. CodeGraph init clobbers config.json on every version; auto-update wiring per §11.4.80 mandate ("ALWAYS latest").
4. cgroup.procs sweep (kill all PIDs not just first) + `stat -f '%z'` portability bug (Darwin vs Linux).
5. Test 09 T4.2 rewritten to verify the REAL containment invariant (cgroup.procs drained), not systemd's unit-state transition (which is sticky on systemd 258 / ALT 11 — verified live).
6. macOS Node 22 LTS pin: codegraph 0.8.0 refuses Node 25 per upstream issue #81. Operator authorised `brew install node@22 && brew link --force --overwrite`. Updated .zshrc references.

Then the hostname-colour palette rebalance:

- Operator: "nezha and Mistborn both show orange". Palette had 7 orange-family colours out of 27.
- Rebalanced to 27 entries spanning the hue spectrum; no two adjacent within RGB Euclidean distance 80.
- NEW test 25: T1 nezha+Mistborn distance ≥ 80 (post-fix: 332.7); T2 16-synthetic-hostname pairwise within pigeonhole tolerance; T3 palette adjacency check.
- NEW M23 mutation: revert palette → test 25 FAILs.

Verification (Darwin quintuple-fresh, 2026-05-21):

- setup.sh --verify-only → PASS=23 FAIL=0 SKIP=2
- meta-test → 34 caught / 0 escaped / 6 skipped
- e2e → PASS=9 FAIL=0 SKIP=0
- codegraph_validate → PASS=4 FAIL=0 SKIP=1
- codegraph_cadence_check → GREEN

Nezha verification: round-5 setup.sh full pipeline GREEN (after fix rounds 1-5 landed). v1.0.7 includes all 6 rounds + palette + test 25.

Out-of-scope this cycle (logged honestly per §11.4.6):

- Auto-detection of Node version compatibility in setup.sh — deferred to v1.0.8.
- Pre-tag Nezha re-verify of v1.0.7 final → will run after tag push as deferred post-release smoke.

§3.12 — v1.0.6 workable-items closure + §11.4.80 cadence + Containers/QWEN.md + release tag (2026-05-21)

Status: COMPLETE (2026-05-21T15:30Z).

Operator directive (verbatim): "Finish all workable items you have presented here, rebuild and install again, retest fully and if fully working as expected release new version!"

Closed (this cycle):

- **§11.4.80 automatic-trigger wiring** (was the one deferred item from v1.0.5 audit). New `scripts/codegraph_cadence_check.sh` + `scripts/ codegraph_install_cadence.sh`. Darwin launchd plist installed + loaded (verified via `launchctl list`). Cross-platform git pre-push hook installed (warns on STALE; `CADENCE_MODE=block` available). Anti-bluff: cadence check reads STAMP CONTENT (regenerated_at + node_count), not just existence. §11.4.81 cross-platform-parity dispatch in the installer: Linux gets systemd user timer; Darwin gets launchd plist.
- **Containers/QWEN.md covenant gap** (was the second-to-last out-of-scope item from v1.0.5 — `Containers/QWEN.md create`). Containers commit `fbef9d6` pushed to github + gitlab. Parent pointer bumped `4ca5491` → `fbef9d6` in this cycle's commit.
- **Full rebuild + install** via `bash scripts/setup.sh` (not just --verify-only). ~/.tmux.conf installed; .bashrc + .zshrc snippets appended; tmx wrapper installed.
- **Quintuple-fresh verification** captured this session: setup.sh --verify-only → PASS=22/0/SKIP=2; meta-test → 32 caught / 0 escaped / 6 skipped; e2e → 9/0/0; codegraph_validate → 4/0/SKIP=1; codegraph_cadence_check → GREEN.

Still out of scope (logged honestly per §11.4.6):

- Cadence-script paired §1.1 mutation (backdate-stamp → cadence FAILs) — deferred to v1.0.7. The script's content-based read is already anti-bluff; the mutation would tighten regression protection.

- CodeGraph upstream `--include-submodules` — separate cycle per §11.4.74.
- Linux-host CI runner — would exercise the Linux systemd user timer install path + the Linux branches of tests 09/13/14.

Release `v1.0.6` tagged + pushed per §11.4.40 (complete fresh retest preceded the tag; no spot-check shortcut).

§3.11 — §11.4.81 cross-platform-parity + Darwin branches + constitution §11.4.81 (2026-05-21)

Status: COMPLETE (2026-05-21T07:30Z).

Triggering events (in arrival order):

- Operator (2026-05-21): "Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform proper implementation will be used for particular OS! EVERYTHING MUST BE PROPERLY EXTENDED AND UPDATED!"
- Operator (2026-05-21): "Add this OS / Platform details / rules / mandatory constraints into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md, QWEN.md and other constitution Submodule relevant files so all projects who inherit these in the future immidiately in such situations create proper missing implementations for particular platform / OS! Make sure you first fetch and pull the latest version of constitution Submodule. Once all done and extended properly commit and push constitution Submodule to all upstreams, then re-process and re-evaluate all rules and mandatory constraints..."
- (Power blackout mid-cycle — work resumed cleanly from working tree.)

Plan + execution captured at `docs/plans/v1.0.5.md`. Seven phases all landed in one v1.0.5 commit per §11.4.42 iteration discipline.

- **§11.4.26 step 1+:** constitution submodule fetched + ff-merged from `7f738df → 19ce1b1` (brought in §11.4.79 + §11.4.80 CodeGraph anchors).

- **§11.4.81 anchor LANDED in constitution submodule (pushed `6e164f3` to `origin` HelixDevelopment).** Universal cross-platform-parity mandate + mirror blocks in Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md. Three sub-mandates: (A) per-OS implementation REQUIRED via runtime dispatch; (B) per-OS tests

REQUIRED with positive captured evidence per branch; (C) honest kernel-gap citation + adjacent equivalent test REQUIRED where no equivalent exists. constitution/QWEN.md ALSO gained the verbatim 2026-04-28 anti-bluff covenant quote (audit gap fix).

- **A26 — §11.4.79 compliance fix.** Removed `constitution/**` and `Containers/**` from `.codegraph/config.json` exclude (own-org MUST be INCLUDED); kept `tmux/**` excluded (third-party). `scripts/codegraph_validate.sh` NEW — 5 probes (V1 CLI version, V2 node count, V3 §11.4.79 split, V4 honest-gap re submodule traversal, V5 MCP spawn). M22 paired mutation (re-exclude → V3 FAILs). Honest gap: CodeGraph 0.6.8 doesn't traverse submodules; config compliance met but practical cross-submodule indexing waits for upstream CodeGraph support.

- **A25 + A22 + A24 — Darwin branches for tests 09/13/14 + NEW test 24 (per §11.4.81 just landed).** All four exercise the macOS POSIX rlimit primitives that are the kernel-enforced equivalents of the Linux cgroup primitives:

    - **Test 09 Darwin (PASS=6/0/0):** wrapper invokes `tmx-rlimit-wrapper.sh`, two operator-path sessions, distinct server PIDs, `ulimit -t`/`-u` readback inside each pane via send-keys + capture-pane, SIGKILL session A's server, session B survives with ORIGINAL PID (positive evidence per §11.4.5).
    - **Test 13 Darwin (PASS=2/0/0):** child bash lowers `ulimit -u 64`
        - fork-bombs; captures EAGAIN occurrences from stderr (`bash: fork: Resource temporarily unavailable`) = XNU kernel-enforced RLIMIT_NPROC.
    - **Test 14 Darwin (PASS=5/0/0):** 3 operator-path sessions; SIGKILL session A's server (macOS adjacent test for OOM-independence per §11.4.81 (C) — Darwin has no OOM killer); verify B+C survive with ORIGINAL PIDs + tmx ls still lists them.
    - **NEW Test 24 Darwin (PASS=2/0/0):** child bash sets `ulimit -t 2`
        - CPU-bound loop; verifies process killed by signal 24 (SIGXCPU) after ~3s wall (exit rc=152 = 128+24); also verifies `TMX_CPU_HARD_SEC=7200` propagates to `RLIMIT_CPU=7200` inside session. §11.4.81 (C) adjacent test for the XNU RLIMIT_AS gap.

- **A25 paired mutations refresh.** M7-M10 RETIRED (targeted dead `scripts/tmx-vm` VM-wrapper path, replaced by native dual-OS per Fixed.md A4-A8). M20 + M21 NEW for Darwin rlimit. M22 for

§11.4.79 codegraph exclude. Meta-test on Darwin: 32 caught / 0 escaped / 6 skipped (M4/M5 topology-correct SKIP on Darwin; M7-M10 retired-with-rationale SKIPs).

- **Constitution submodule pointer bumped** `19ce1b1 → 6e164f3` in this same project commit per §11.4.26 step 7.

Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --verify-only` PASS=22/0/SKIP=2, meta-test 32/0/SKIP=6 (all SKIPs §11.4.3 topology-correct or retired-with-rationale), e2e 9/0/0, codegraph_validate PASS=4/0/SKIP=1.

Out-of-scope this cycle (honest tracking per §11.4.6):

- §11.4.80 automatic-trigger wiring (cron / git hook for the constitution-provided `codegraph_update.sh` + `codegraph_sync.sh`) — deferred. Manual invocation works today.
- CodeGraph upstream `--include-submodules` support — out of scope per §11.4.74.
- `Containers/QWEN.md` create — separate PR to `vasic-digital/Containers` per §11.4.28.
- Linux-host CI runner — would let us exercise the Linux branches of tests 09/13/14 alongside the Darwin branches running today.

§3.10 — CodeGraph (§11.4.78) + verbatim covenant + AUDIT fixes + docs reorg (2026-05-21)

Status: COMPLETE (2026-05-21T05:30Z).

Triggering events (in arrival order):

- Operator (2026-05-21): create full working plan for tackling all these points completely; all existing tests + Challenges MUST work anti-bluff; covenant MUST be part of project + submodule Constitution / CLAUDE / AGENTS; HelixConstitution submodule is the root.
- Operator (2026-05-21): incorporate / install CodeGraph for Claude Code, OpenCode, Kimi CLI, Crush, Qwen Code; comprehensive anti-bluff tests.
- Operator (2026-05-21): full documentation + user guides + HTML + PDF exports.
- Operator (2026-05-21): do NOT modify constitution submodule, only keep it regularly updated.
- Operator (2026-05-21): documentation under context-named subdirectories per the constitution rule.

Plan + execution captured at `docs/plans/v1.0.4.md`. Eight phases all landed in one v1.0.4 commit per §11.4.42 iteration discipline.

- **A18 CodeGraph integration (§11.4.78).** CLI v0.6.8 installed; `.codegraph/config.json` tracked with §11.4.10 secret exclusions + §11.4.28 owned-submodule paths; `.codegraph/codegraph.db` gitignored; §11.4.77 regen manifest + `scripts/codegraph_reindex.sh`. MCP wired for all 5 agents (Claude Code `.mcp.json` NEW; OpenCode + Kimi audited; Crush `.crush.json` NEW; Qwen Code `.qwen/settings.json` NEW). All configs reference bare `codegraph` on PATH. Comprehensive `docs/codegraph/README.md`. Tests 20/21/22, challenges CH-20/21/22, mutations M16/M17.

- **A19 verbatim covenant propagation.** §11.4 2026-04-28 user mandate now LITERALLY present in project `CLAUDE.md` + `AGENTS.md` + `QWEN.md` (was only in `Constitution.md`). Verify.sh Layer-1 gate added; test 19 PASS=7/0/0; CH-19; M15 (mutates a temp copy — real CLAUDE.md is never touched).

- **A20 AUDIT-1 fix.** M4/M5 meta-test mutations were silently SKIPping on Darwin for the wrong reason (BSD `sed -i` quirk). Real root cause: the mutations target Linux-only cgroup/systemd-run wrapper code unreachable on Darwin. Fix: explicit `uname -s` topology guard per §11.4.3 + portable `inplace_sed` for Linux runs.

- **A21 AUDIT-2 fix.** `tmx kill -t NAME` (documented friendly verb) was ambiguous because tmux only knows `kill-pane`/`kill-server`/ `kill-session`/`kill-window`. Wrapper now translates the bare `kill` verb to `kill-session`. Test 23 PASS=5/0/0; CH-23; M19.

- **Docs reorganised** under context-named subdirectories per the operator's invocation of the constitution rule: 7 docs moved (GUIDE/SCROLLING/CODEGRAPH/CONTAINERIZATION_PLAN/ NATIVE_DUAL_OS_PLAN/ PER_SESSION_ISOLATION_PLAN/ PLAN_v1.0.4). Every reference in governance + scripts updated atomically.

- **§11.4.65 universal-Markdown export.** New `scripts/export_docs.sh` (pandoc HTML + weasyprint PDF, idempotent, per-file timeout 60s) refreshed siblings for every consumer Markdown.

- **§11.4.71 pre-push.** Parent + `constitution/` + `Containers/` all at upstream tip; no divergent commits.

- **Honest gaps logged (§11.4.6) — out of scope this cycle:**
  - Upstream `constitution/QWEN.md` covenant insert → separate PR to `HelixDevelopment/HelixConstitution` (operator forbids modifying constitution from inside this project).
  - `Containers/QWEN.md` create + populate → separate PR to `vasic-digital/Containers` per §11.4.28.
  - Shell parser for CodeGraph (currently only the C file indexed, 6 nodes) → upstream contribution to add tree-sitter shell.
  - Agent-driven unforgeable-challenge end-to-end test → classified `AUTONOMOUS_DESIGNED` per §11.4.52 carve-out (mechanical seam exists via test 22 T7; agent-driven layer lands when a headless agent harness is wired).

Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --verify-only` PASS=18/0/SKIP=5, meta-test 26/0/SKIP=6 (all SKIPs §11.4.3 topology-correct), e2e 9/0/0.

§3.9 — Scrolling fix + HelixConstitution inheritance (2026-05-21)

Status: COMPLETE (2026-05-21).

Triggering event: operator research note — tmux scrolling (up/down of terminal output, especially inside the Claude Code TUI) did not work; requirement to scroll vertically from ANY computer or mobile phone (Termux/Android). Plus the mandate to incorporate the HelixConstitution submodule as the root of the Constitution / CLAUDE.md / AGENTS.md, inherited further.

- **A16 scrolling fix** (`Fixed.md` A16): `scripts/tmux.conf.template` — history-limit 2000→50000, `mode-keys vi`, `WheelUp`/`WheelDown` overridden to always drive copy-mode scrollback (works on desktop mouse + trackpad + Termux touch-scroll), `allow-passthrough` + `extended-keys` + `terminal-features extkeys` for the Claude Code TUI, OS-adaptive `@clip` clipboard. NEW test 17 (operator-path, PASS=13/0/0), verify.sh Layer-1 static gate, `TMUX-CH-17`, M12 + M13 paired mutations.
- **A17 HelixConstitution** (`Fixed.md` A17): added the `constitution/` submodule (pinned `7f738df`). Full refactor of `Constitution.md` to the extends-template form (Project Articles §101–§109); `CLAUDE.md` / `AGENTS.md` inheritance pointers; new `QWEN.md`. NEW test 18 (PASS=10/0/0), `TMUX-CH-18`, M14 + `CM-CONSTITUTION-INHERITANCE` paired mutations. The constitution mutation runs on a TEMP COPY

— the real, decoupled `constitution/` submodule is never modified (operator directive).

- **Containers submodule**: adopted remote `4ca5491`, which already carries HelixConstitution recursive inheritance (`find_constitution.sh`, `QWEN.md`, all four governance docs). Parent gitlink bumped `b077f2c → 4ca5491`.
- **Meta-test portability**: M1/M2/M3/M6 converted from GNU `sed -i` to a portable `inplace_sed` helper — they now run on macOS instead of silently skipping. Meta-test: 18 caught / 0 escaped / 6 skipped (was 10/0/10).
- Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --rebuild` GREEN, suite PASS=13/0/SKIP=5, meta-test 18/0/6, e2e 9/0/0.

§4 — Recent commits

- (this commit, 2026-05-21) — **A16 scrolling fix + A17 HelixConstitution inheritance (v1.0.3 / versionCode 4).** `tmux.conf.template`: history-limit 50000, vi copy-mode, WheelUp/Down → copy-mode scrollback override, Claude Code passthrough, OS-adaptive clipboard. HelixConstitution submodule added at `constitution/`; `Constitution.md` refactored to the extends-template form (§101–§109); CLAUDE/AGENTS inheritance pointers + new `QWEN.md`. Tests 17 + 18, challenges CH-17/18, mutations M12–M14 + CM-CONSTITUTION-INHERITANCE; meta-test M1/M2/M3/M6 made portable. Containers gitlink → `4ca5491`. Gate: PASS=13/0/5, meta 18/0/6, e2e 9/0/0. See `Fixed.md` A16 + A17, §3.9.
- (this commit, 2026-05-16) — **A15 cosmetic .exe-strip cycle (v1.0.2 / versionCode 3).** Bottom-left status bar showed `claude.exe` because Claude Code v2.x ships its macOS native binary literally as `.../@anthropic-ai/claude-code/bin/claude.exe` (verified Mach-O 64-bit ARM64). Patched `scripts/tmux.conf.template` to set `automatic-rename-format "#{s/\\.exe$//:pane_current_command}"` — literal-dot-anchored, escape-verified-empirically. Added test 16 (operator-path per §11.4.7, includes regression guard binary `t16_bashexe` to catch the unescaped-dot bug class), challenge TMUX-CH-16, M11 paired mutation. Live operator-path validation recorded: `pane_current_command='claude.exe' → #W='claude'`. Full gate: PASS=11 FAIL=0 SKIP=5 (same SKIP profile as v1.0.0). See Fixed.md A15.
- (commit before that, 2026-05-13) — Audit cycle: tmux submodule reverted from `3f651d9f` to `cc117b5` (3.6a tag) after silent pin-drift caught; README.md test-count + PASS/SKIP staleness fixed (8→14 tests, PASS=6→10, SKIP=2→4); CLAUDE.md + AGENTS.md gained

project summary, cross-links, fresh-conversation workflow, `NN_*.sh` glob fix, named four layers, "Files to never edit directly"; Issues.md restored A/B/C/D/E section headers; CONTINUATION.md §3.8 + timestamp refresh per §12.10. See §3.8.
- `f4132aa` — Auto-commit: opaque submodule bumps (Containers `e377dea→af51968` legitimate upstream governance; tmux `cc117b5→3f651d9f` reverted in this commit). §12.10 / §11.4.6 violation documented in §3.8.
- (previous commit) — §11.4.4 layer-4 paired-mutation harness (META-MUT-001): meta_test_false_positive_proof.sh with 5 mutations (10/0/0 all caught); CLAUDE.md/AGENTS.md PENDING→LANDED; Issues.md now empty of open items. Also: fixed 5 broken script references (REPO_ROOT wrong depth, filenames); implemented systemd-run --user --scope wrapping in tmx.template; test 09 topology dispatch fixed (functional probe, not mount grep); CLAUDE.md compacted to match AGENTS.md; binary built and verified GREEN (PASS=10 FAIL=0 SKIP=4: OOM helper + 3 destructive).
- `39284ab` — Full coverage: CH-09 added (was missing); tests 12/13/14 (T5/T7/T8 destructive) with TMX_TEST_DESTRUCTIVE=1 gate; topology probe in tmx.template; challenge entries CH-12/13/14; B1/C1/C2/C3/D1 migrated Issues→Fixed.
- `8b37046` — Hostname-derived status-bar colour: DJB2→27-colour palette algorithm + wrapper integration + tests 10/11 + challenges + docs.
- `68a65b0` — Anti-bluff enforcement: TEST-AUDIT-001 complete (9 tests §11.4.2-audited), challenges paths fixed, AGENTS.md compact rewrite, covenant propagation verified.
- `b92bf7f` (2026-05-08) — §1 anti-bluff covenant verbatim user-mandate quote added; test 09 crash-isolation-scope (14/0/0) landed.
- `08d4ba5` (2026-05-07) — Initial vasic-digital/tmux: migrated from ATMOSphere project + per-session containerization plan + 8-test verification gate.

§8 — Resumption recipe

1. `cd ~/Projects/tmux`
2. Read this document
3. Read `Constitution.md`, `CLAUDE.md`, `AGENTS.md`, `Issues.md`, `Fixed.md`
4. Find the topmost IN PROGRESS / OPEN item, resume
5. Update this document in the SAME commit as the work itself (§12.10 invariant)